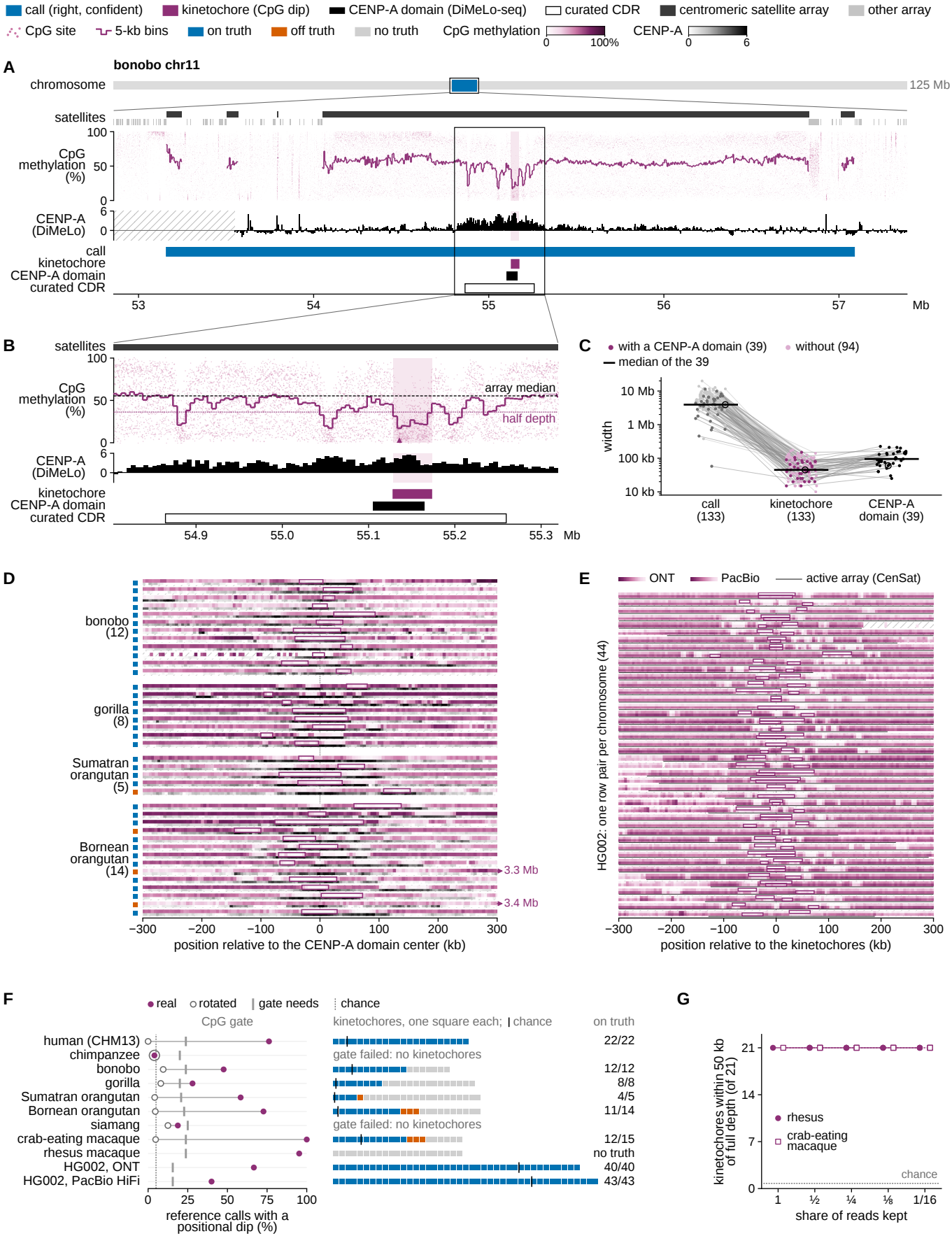

call (right, confident)
kinetochore (CpG dip)
CENP-A domain (DiMeLo-seq)
curated CDR
centromeric satellite array
other array
CpG site
5-kb bins
on truth
off truth
no truth
CpG methylation
0
100%
CENP-A
0
6
A
bonobo chr11
chromosome
125 Mb
satellites
CpG methylation (%)
CENP-A (DiMeLo)
call
kinetochore
CENP-A domain
curated CDR
53
54
55
56
57
Mb
B
array median
half depth
54.9
55.0
55.1
55.2
55.3
Mb
C
with a CENP-A domain (39)
without (94)
median of the 39
width
10 Mb
1 Mb
100 kb
10 kb
call (133)
kinetochore (133)
CENP-A domain (39)
D
bonobo (12)
gorilla (8)
Sumatran orangutan (5)
Bornean orangutan (14)
3.3 Mb
3.4 Mb
position relative to the CENP-A domain center (kb)
E
ONT
PacBio
active array (CenSat)
HG002: one row pair per chromosome (44)
position relative to the kinetochores (kb)
F
real
rotated
gate needs
chance
CpG gate
kinetochores, one square each; | chance
on truth
human (CHM13)
22/22
chimpanzee
gate failed: no kinetochores
bonobo
12/12
gorilla
8/8
Sumatran orangutan
4/5
Bornean orangutan
11/14
siamang
gate failed: no kinetochores
crab-eating macaque
12/15
rhesus macaque
no truth
HG002, ONT
40/40
HG002, PacBio HiFi
43/43
reference calls with a positional dip (%)
G
kinetochores within 50 kb of full depth (of 21)
rhesus
crab-eating macaque
chance
share of reads kept

In primates the kinetochore sits in a CpG-hypomethylated dip tens of kilobases wide inside the megabase-long satellite array, so the methylation calls of ordinary long reads locate CENP-A chromatin. Centromere identity leaves a mark that any long-read primate genome carries.

A A typical bonobo centromere. The satellite array spans megabases and its CpGs are about half methylated; in one place methylation drops, and that is where CENP-A, the histone that marks the kinetochore, is highest.

B The same place at the scale of single CpGs: the hypomethylated dip is tens of kilobases wide and sits inside the CENP-A domain, among shallower dips of the same array.

C The dip narrows the centromere from the megabase array to tens of kilobases, about half the width of the CENP-A domain itself.

D Every ape kinetochore read out this way, aligned on its CENP-A domain: the light dip and the dark CENP-A signal line up on nearly every chromosome, in four species.

E The same diploid human genome read by two sequencing technologies gives dips in the same places.

F The dips sit at centromeres far more often than methylation moved from elsewhere on the chromosome would put them, in most primates; where the dips fail this test (chimpanzee, siamang), no kinetochore is placed. Wherever truth exists, most kinetochores land on it.

G A sixteenth of the reads, about one methylation call per CpG, still places every kinetochore in the same place.

Terms. cenfuse 0.10.1 with FASTA, genes, Hi-C and the public ONT CpG track of each assembled animal (apes R9, rhesus R10; HG002: FASTA and CpG only). Kinetochore: the most prominent valley of CpG methylation inside the call (5-kb bins over the arrays of the centromeric satellite family), extended to the bins below half its depth; reported where the genome's CpG gate passes and the dip beats the call's own bins shuffled ($P \le 0.05$). CENP-A domain: half-maximum run of DiMeLo-seq CENP-A signal (antibody-directed adenine methylation, independent of the CpG calls). On truth: within 50 kb of the CENP-A domain (apes), overlapping the curated centromere dip region (CDR; human), wholly inside the CENP-A ChIP domain (crab-eating macaque) or an active alpha-satellite array (HG002). Chance: a kinetochore of the same width placed at random inside its call.

A Bonobo chr11 (CM055483.2), chosen by rule: of the 39 ape kinetochores with a CENP-A domain in their call, the one nearest the medians of kinetochore width (45 kb) and of the distance between kinetochore and CENP-A centers (23 kb) (next: gorilla CM068951.1, bonobo CM055476.2). Its call (3.93 Mb, confidence high) is the median of the 39. Top: the whole chromosome (125 Mb). Tracks: every satellite array (dark: the centromeric family, a 171-bp alpha satellite whose largest array is 2.76 Mb; light: others); every CpG of the input track (56,225 dots here) and the 5-kb bins cenfuse used (line); DiMeLo-seq CENP-A (hatched: no data); the call (blue: right, confident), the kinetochore (55.128-55.173 Mb), the CENP-A domain and the curated CDR. Pink band: the kinetochore. The array is methylated at 55% of its CpGs, the kinetochore at 24%.

B The boxed window of A. Dashed: array median of the 5-kb bins; dotted: half depth; triangle: the valley floor. The kinetochore (45 kb) overlaps the CENP-A domain (60 kb; centers 16 kb apart). The curated CDR (395 kb) spans several dips of similar depth; cenfuse reports the most prominent. Dip test $P = 0.005$ (200 shuffles), positional $P = 0.025$ (200 rotations).

C Widths, one point per kinetochore; lines join a chromosome's call, kinetochore and CENP-A domain; circles: the example. Dark: the 39 ape kinetochores with a DiMeLo-seq CENP-A domain in their call; bars: their medians, kinetochores 45 kb, calls 3.9 Mb (87× narrower), CENP-A domains 95 kb. Light: the other 94 kinetochores of the six genomes with a passing gate and truth (human 22, crab-eating macaque 21, bonobo 7, gorilla 15, Sumatran orangutan 19, Bornean orangutan 10). All 133 together: kinetochores 55 kb, calls 5.1 Mb (92×).

D One kinetochore per row pair: CpG methylation per 5-kb bin (upper strip; hatched: fewer than 10 CpGs) and DiMeLo-seq CENP-A (lower strip; clipped at 6), centered on the CENP-A domain, ± 300 kb; box: kinetochore; arrow: kinetochore outside the window; square: within 50 kb of CENP-A (blue) or not (vermilion); rows by chromosome. 35 of 39 within 50 kb (29 overlap; chance 4.9, Poisson-binomial $P = 1.9 \times 10^{-34}$); per species: bonobo 12/12, gorilla 8/8, Sumatran orangutan 4/5, Bornean orangutan 11/14. CENP-A signal is higher inside the kinetochore than over its call on 39/39 (median 5.02 vs 0.81). Misses: two in the same array, Bornean orangutan CM054635.2 (50 kb, 1 bp beyond 50), Sumatran orangutan CM054703.2 (65 kb); two Bornean orangutan kinetochores lie in the large 171-bp array 3.3 and 3.4 Mb from CENP-A, which sits on small arrays of 181- and 189-bp families inside the same call. Another 7 ape calls hold a CENP-A domain but no significant dip ($P > 0.05$), so no kinetochore. Chimpanzee has none: its gate failed.

E HG002 (diploid T2T, both haplotypes) with CpG methylation from ONT R10 (upper strip) and PacBio HiFi (lower strip) of the same cell line; boxes: each technology's kinetochore; line: CenSat active alpha-satellite arrays; rows centered on the mean of the two kinetochore midpoints, chr1 to chrY, maternal first (44 chromosomes with a kinetochore from either). The gate passed on 46 chromosomes with each. Kinetochores: ONT 40 (median 38 kb), PacBio 43 (25 kb), all wholly inside an active array (chance 30 and 32: calls lie on the arrays). Where both place one (39), they are within 50 kb on 31 (overlap 15); the other 8 lie 55-195 kb apart in the same array. Calls equal the FASTA-only run's. HG002 has no CENP-A data.

F Left, the CpG gate. A dip is positional when it is longer than dips made by the chromosome's own CpG values rotated onto the call ($P \le 0.05$, 200 rotations). The gate passes when the other chromosomes' high-confidence calls hold more positional dips than 5% each would give (binomial $P \le 0.01$; gray tick: dips needed). Filled: real tracks; open: the same calls and CpG line with every value moved half a chromosome along the same CpGs (lines join the pair). Real: passes in 6/7 primates with truth, rhesus and HG002; fails in chimpanzee (1 of 25) and siamang (3 of 16). Rotated: 0-2 positional dips among the references; the gate passes on 0 of 197 chromosomes (0/8 genomes). Right: one square per kinetochore; black tick: chance. Human truth is the curated CDR, from the same reads (an implementation check); with the apes 62/63 overlap their CDR (Sumatran orangutan CM054694.2 is 2.3 Mb off). Crab-eating macaque: 12/15 inside the CENP-A ChIP domain of another animal; the other three lie 0.8-3.4 Mb from it inside the same call. Rhesus (held out of the design): 21 of 22 chromosomes with a kinetochore.

G The two tables with read counts (rhesus R10, about 15 calls per CpG strand; crab-eating macaque about 38) thinned binomially to 1/2 to 1/16 of the reads, 3 seeds each; kinetochores within 50 kb of the full-depth run's. These runs predate 0.10.1 (2026-10-27) and use its CpG rule; their calls differ from 0.10.1's on 3 chromosomes, and their full-depth kinetochores lie within 50 kb of 0.10.1's on 42/42. Rhesus at 1/16 (about one call per CpG strand): 21/21 in every seed; crab-eating macaque 21/21; chance at most 0.8 (dotted). No call changed.

Exceptions and caveats. The CpG line refines a call and never moves it or changes its confidence (calls with and without it are identical on all 311 chromosomes). A wrong call gets a kinetochore inside the wrong array (Bornean orangutan CM054652.2, the confident ape miss, gets one in its inactive array, methylation 0.15 vs 0.32; absent from D, whose calls hold CENP-A). A kinetochore does not depend on the call's confidence: Bornean orangutan CM054644.2, whose call is low (a competing satellite cluster that Hi-C cannot tell apart), has its kinetochore on CENP-A. The CpG rules were designed on these primate tracks (not blind); rhesus and HG002 were held out. The example is typical of all 39 by the rule above, not the cleanest: its CDR holds several dips and the kinetochore is one of them.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/figures/benchmark_cpg/benchmark_cpg.py · 2026-11-04